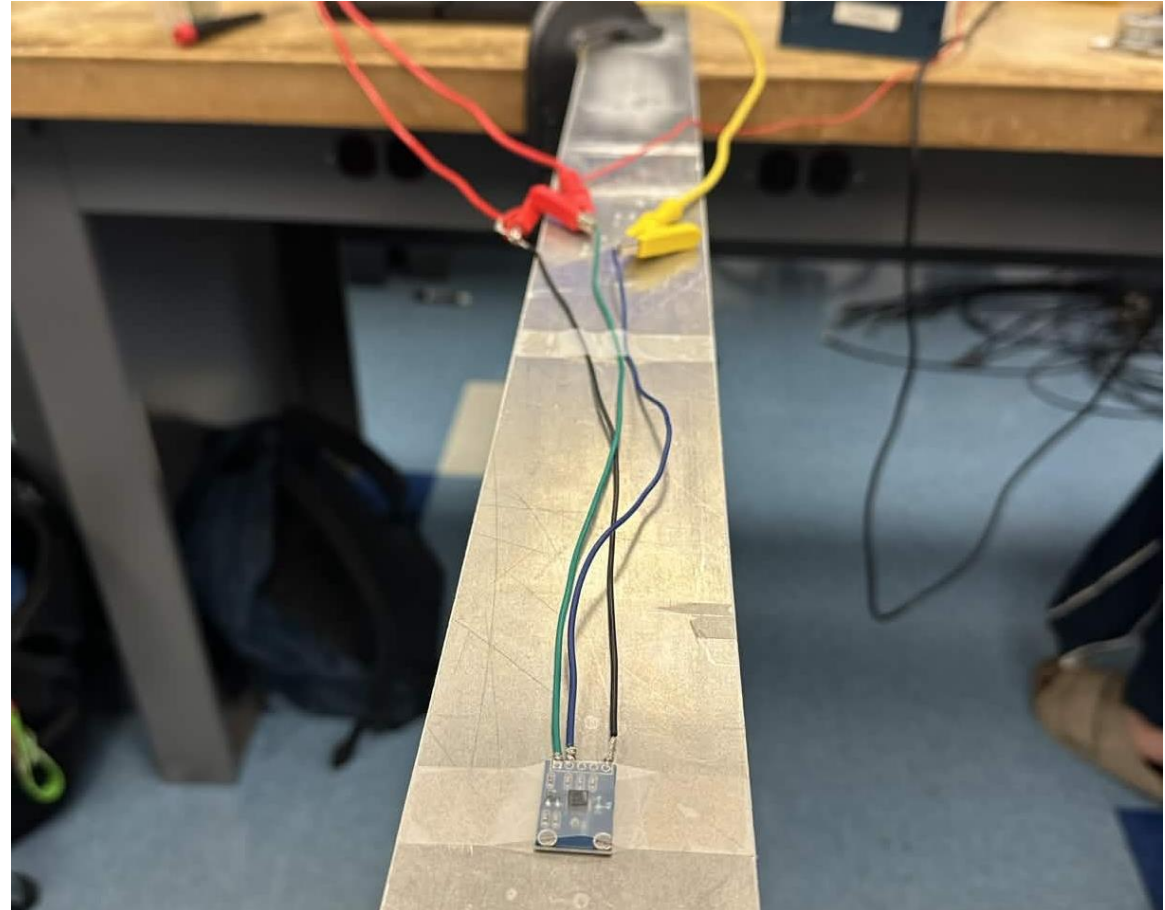


Cantilever Beam Vibration Project



Objective and Expected Behavior

Objective

Measure the cantilever beam response after excitation and compare how the period and frequency change when 1 lb, 3 lbs, and 5 lbs are added.

Expected trend

A larger attached mass increases the effective inertia of the beam system, so the oscillation period should increase and the dominant frequency should decrease.

Key ideas used in the analysis

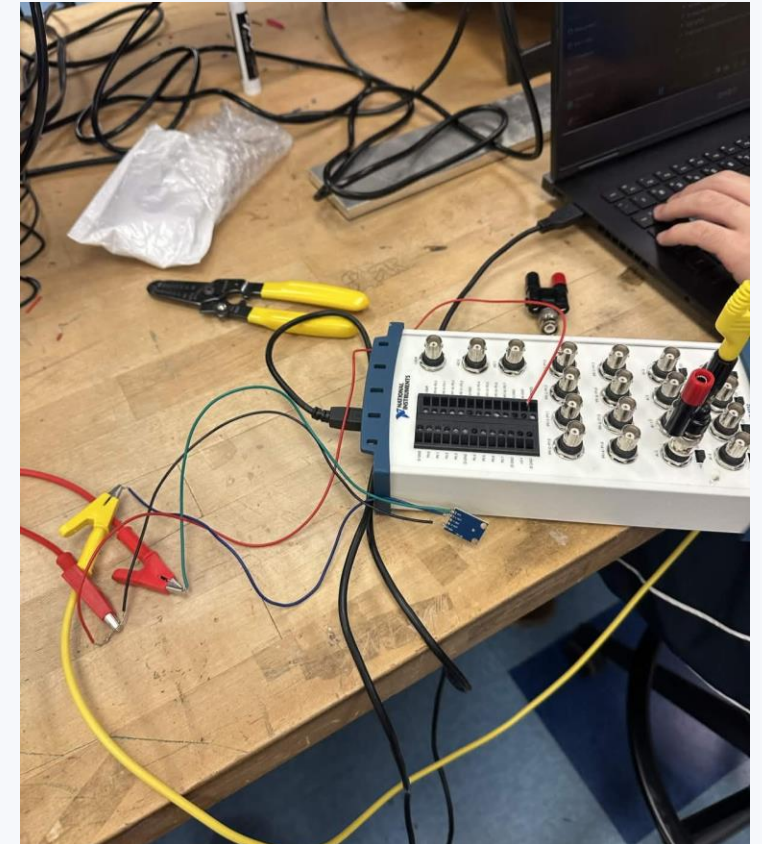
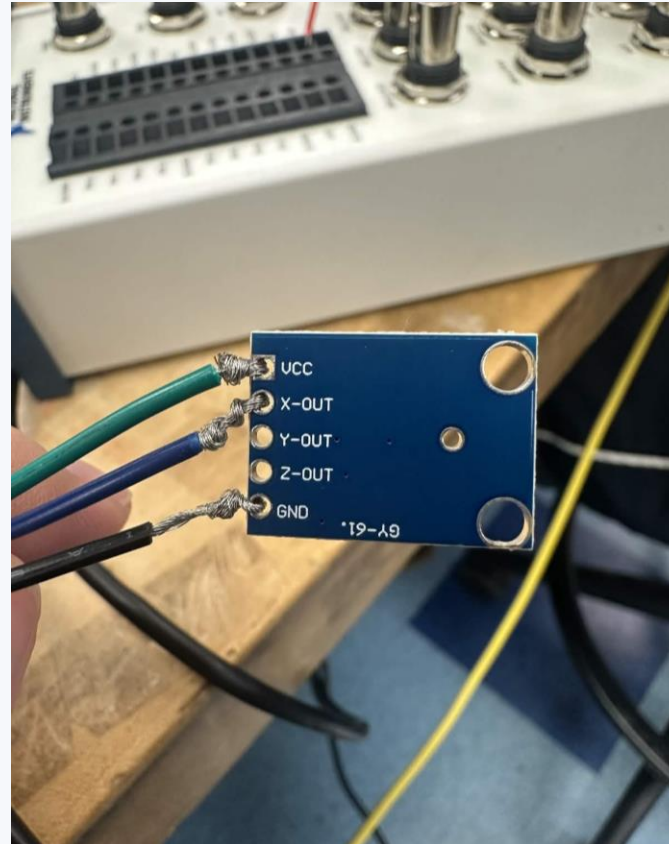
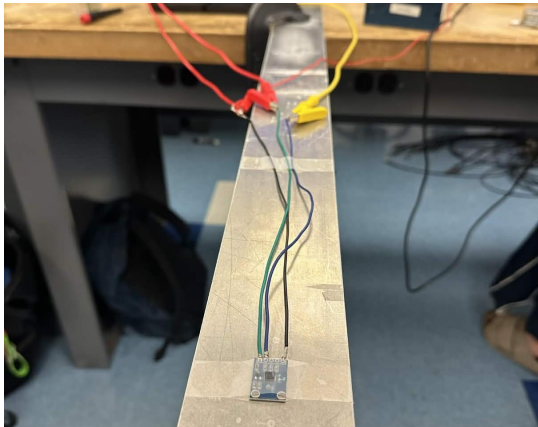
- The beam was excited and then allowed to vibrate freely.
- The accelerometer output was recorded as voltage versus time.
- The time between consecutive peaks was used to estimate the period T .
- Frequency was then calculated from the measured period using $f = 1/T$.
- The results were compared across no mass, 1 lb, 3 lb, and 5 lb.

$$\text{Frequency equation: } f = 1/T$$

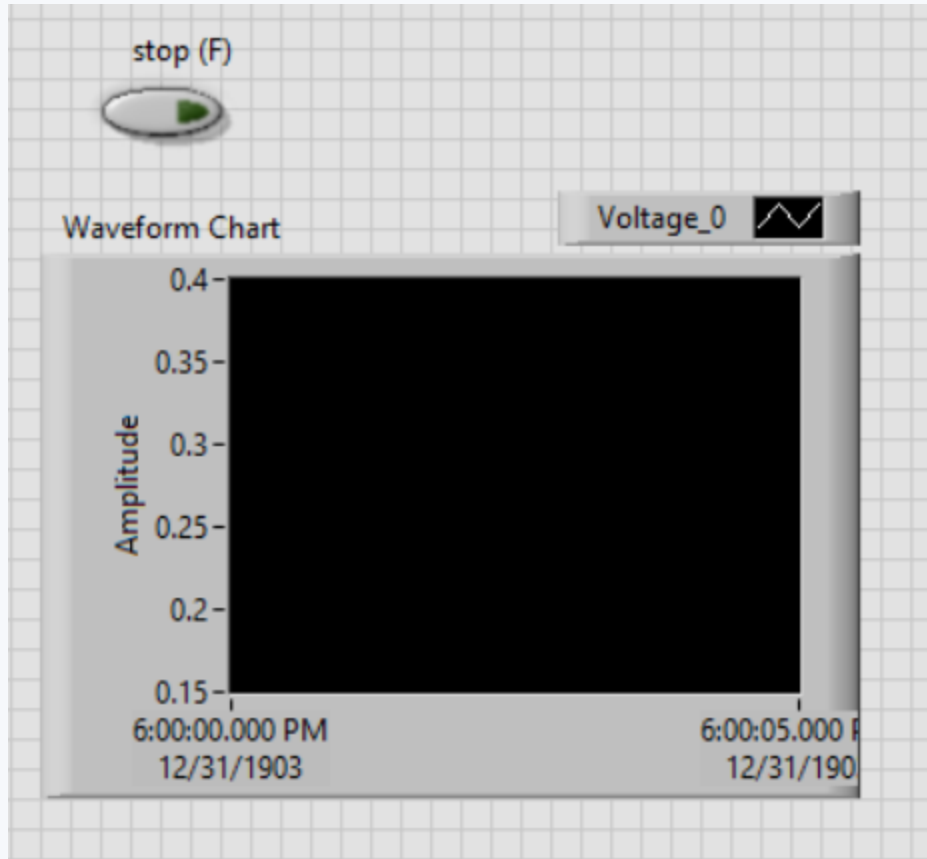
Equipment Used

Main equipment

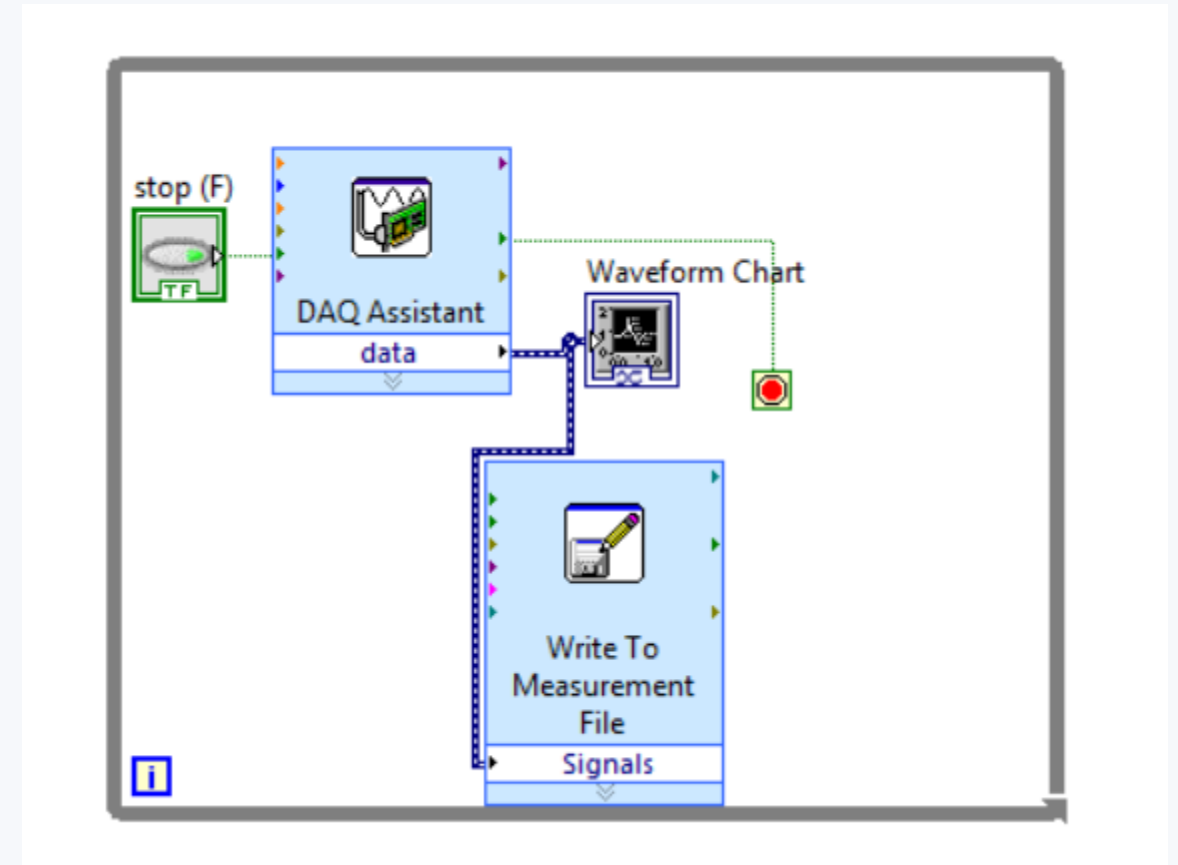
- Metal cantilever beam
- Accelerometer mounted near the free end
- National Instruments USB-6341 DAQ
- Computer running LabVIEW
- Connecting wires and clips
- Added masses: 1 lb, 3 lbs, and 5 lbs
- Power Bank for 3.3V



LabVIEW Setup

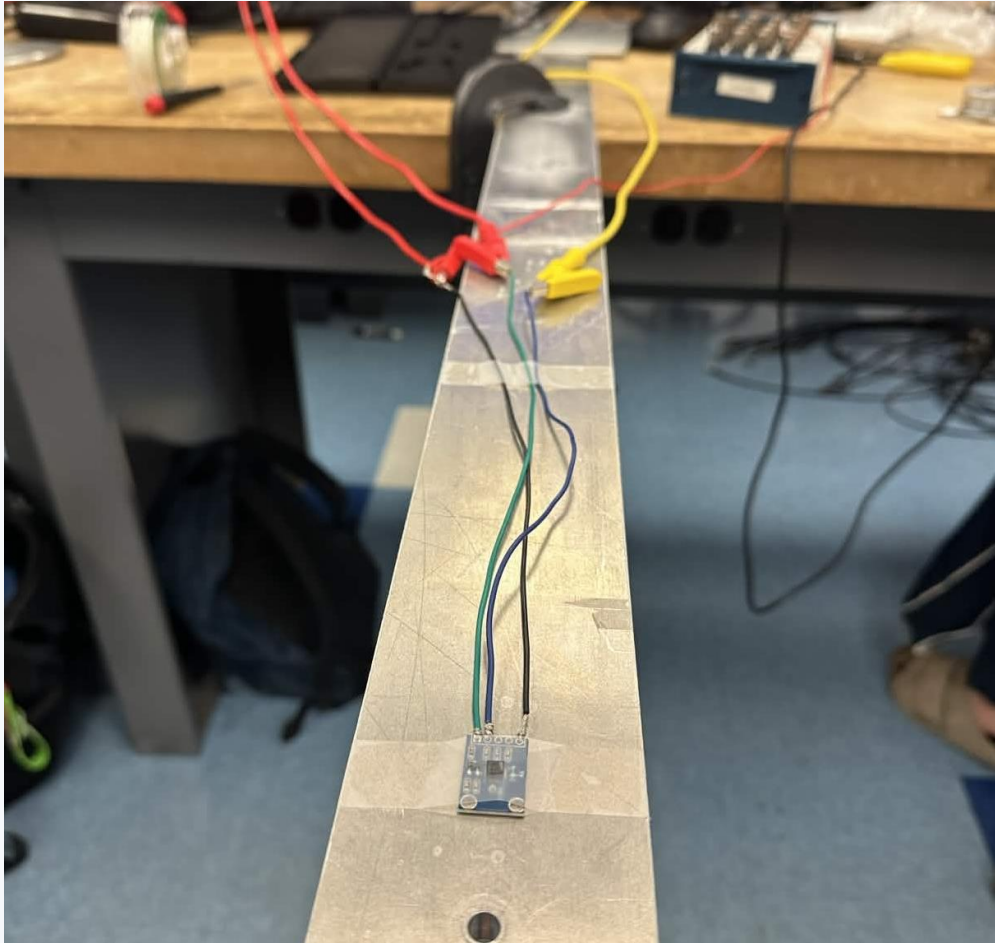


Front Panel of vi

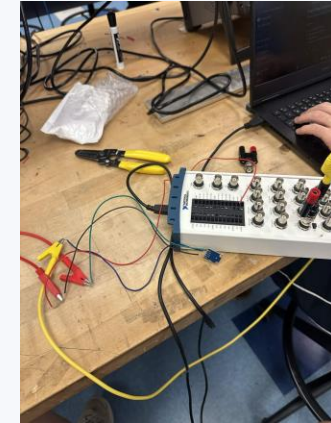
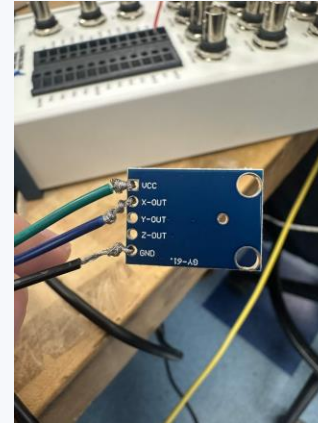


Block Diagram of vi

Experimental Setup



Beam with accelerometer attached



How the system worked

The accelerometer converted the beam motion into a voltage signal. The DAQ sent that signal into LabVIEW, where each trial was recorded as a time-based measurement file.

Procedure and Data Collection

Step 1

Mount the accelerometer on the beam and connect VCC, signal output, and ground to the DAQ.

Step 2

Confirm that LabVIEW is reading a stable signal before exciting the beam.

Step 3

Excite the beam by tapping and record the response as voltage versus time.

Step 4

Repeat for no mass, 1 lb, 3 lb, and 5 lb. Save each trial as an LVM file.

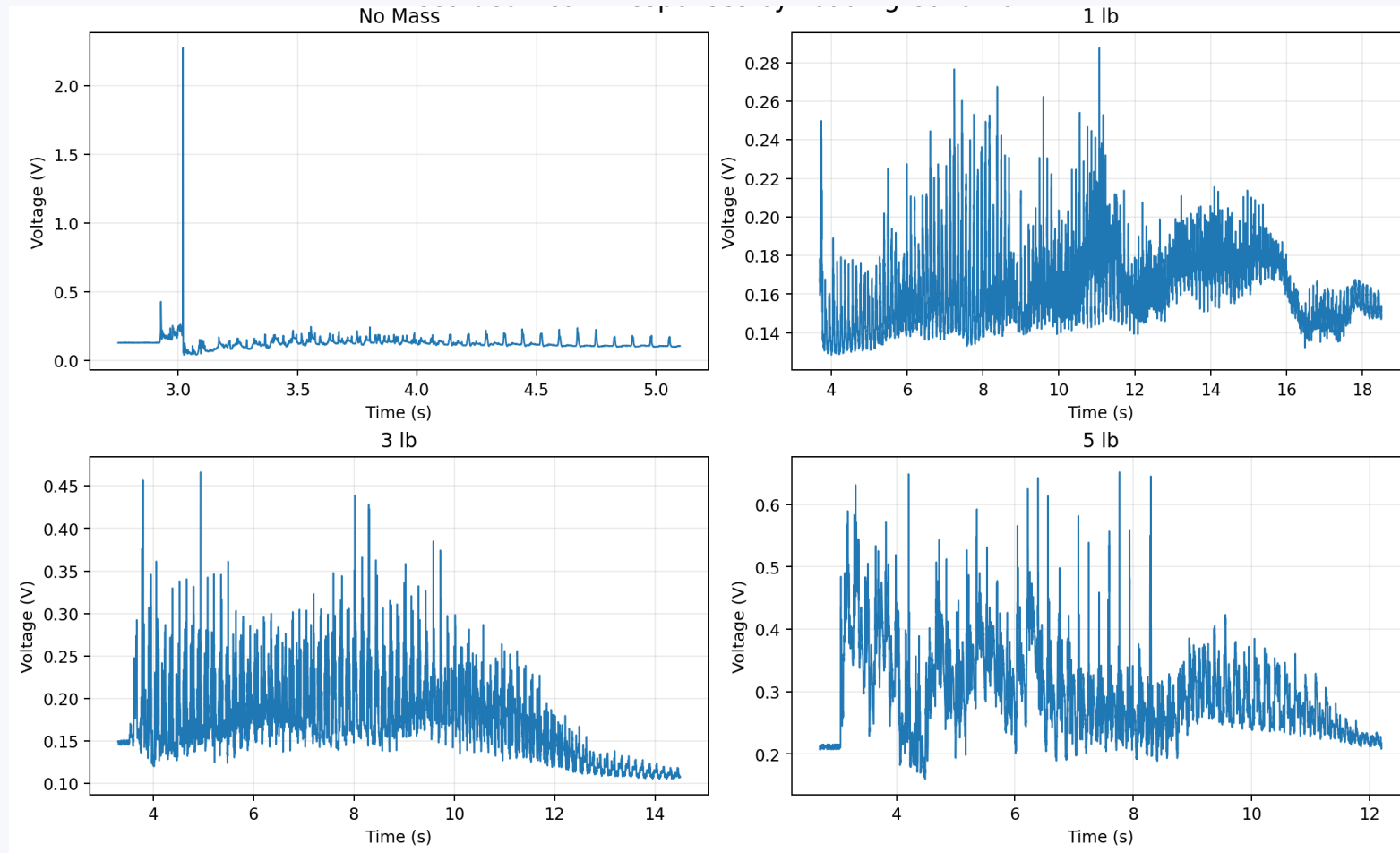
Step 5

Use the recorded peaks to estimate period and then calculate frequency.

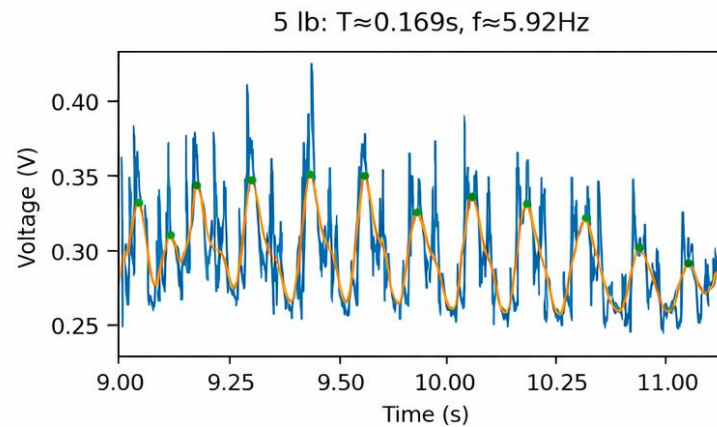
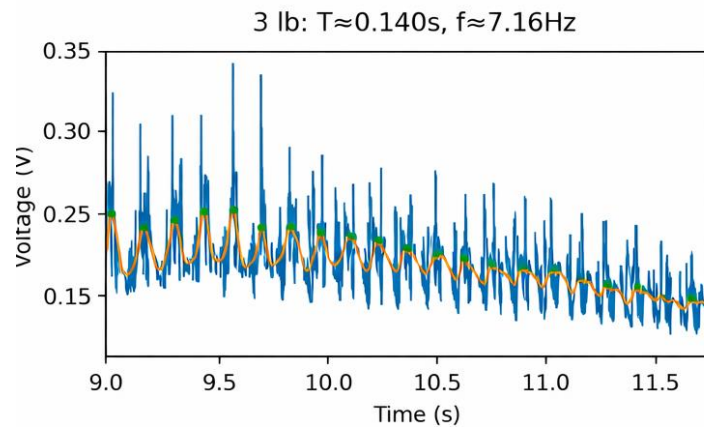
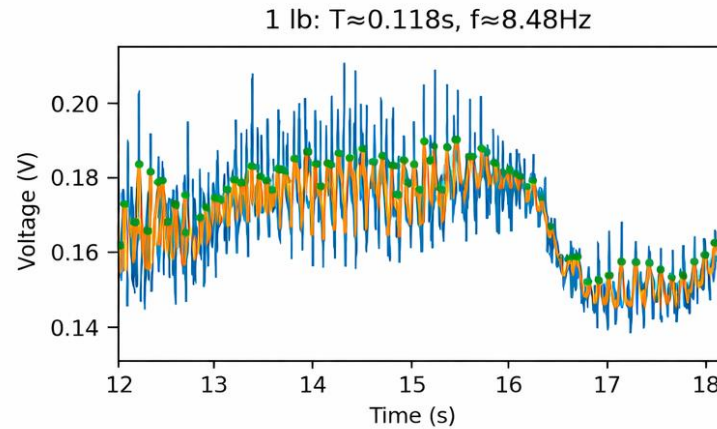
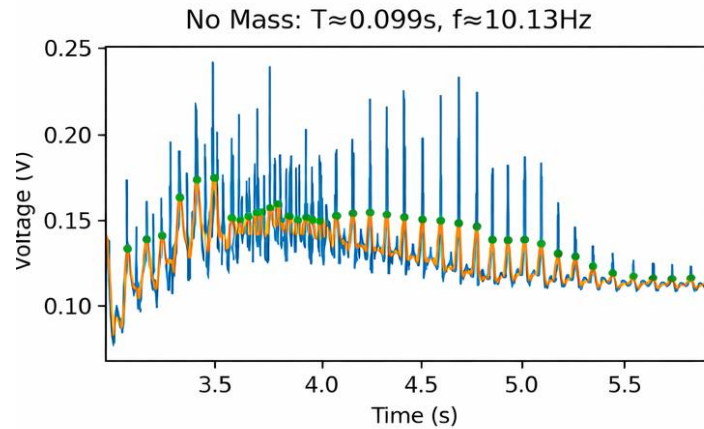
Sampling details

The LabVIEW files used a sampling interval of 0.001 s, which corresponds to a 1000 Hz sampling rate. That gave enough time resolution to identify the main oscillation peaks and compare the four loading conditions.

Raw Recorded Responses



Estimating Period and Frequency from Recorded Peaks



Peak-Based Calculation

For each trial:
 $T \approx$ average time
between consecutive
peaks
 $f = 1/T$

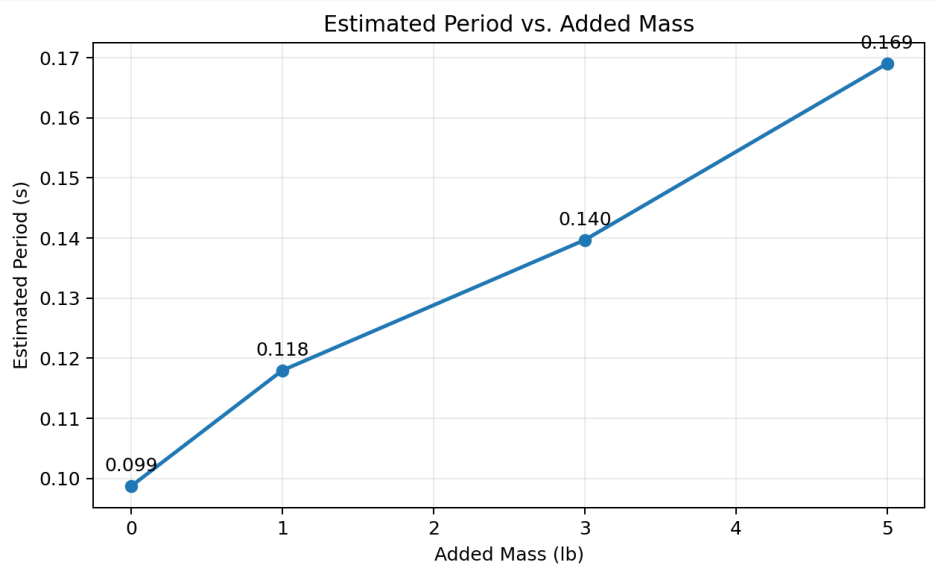
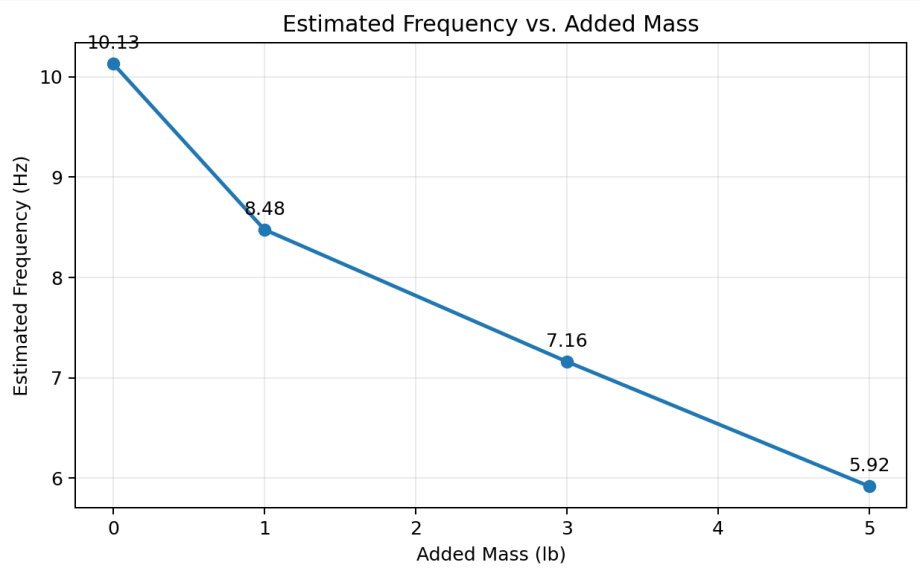
Condition	T (s)	f (Hz)
No Mass	0.099	10.13
1 lb	0.118	8.48
3 lb	0.140	7.14
5 lb	0.169	5.92

These estimates were taken directly from the processed measured peaks in the selected post-excitation windows.

Results

The period increased and the frequency decreased as more mass was added.

Condition	Mass (lb)	Estimated Period (s)	Estimated Frequency
No Mass	0	0.099	10.13
1 lb	1	0.118	8.48
3 lb	3	0.140	7.16
5 lb	5	0.169	5.92



What the Results Mean

Frequency trend

Estimated frequency dropped from about 10.13 Hz with no mass to about 5.92 Hz at 5 lb. That means the beam oscillated more slowly as load increased.

Period trend

Estimated period increased from about 0.099 s with no mass to about 0.169 s at 5 lb, which means each cycle took longer with heavier loading.

Overall answer

Added mass clearly changed the beam response. Based on the processed peaks, more mass caused a lower dominant frequency and a longer oscillation period.

Our measurements support the expected conclusion for a cantilever beam: adding mass slows the vibration response. The exact values are specific to our setup, but the overall trend is consistent with beam vibration theory.

Limitations and Sources of Error

- Some trials contained electrical noise and abrupt spikes, which made the signals less clean than an ideal damped sinusoid.
- The tap used to excite the beam was not perfectly identical for every trial, so the starting amplitude varied.
- The frequency values are estimates based on recorded peaks, so they should be interpreted as approximate experimental results rather than exact theoretical values.

Implications

- Cantilever beam vibration is important in real engineering systems because added mass changes how fast a structure vibrates.
- Engineers use this idea to predict vibration behavior and avoid unwanted resonance.
 - Buildings and bridges: extra load can change natural frequency and affect structural response.
 - Airplane wings: attached equipment or payload changes vibration behavior.
- Our results show that increasing mass lowers frequency and increases period, which is useful for designing safer and more reliable structures.

Conclusion

Our experiment showed that adding mass to the cantilever beam made it vibrate more slowly. From the recorded data, the estimated period increased and the estimated frequency decreased as the load changed from no mass to 5 lb. This trend agrees with the expected behavior of a cantilever beam and shows that added mass has a clear effect on vibration response.

Questions?